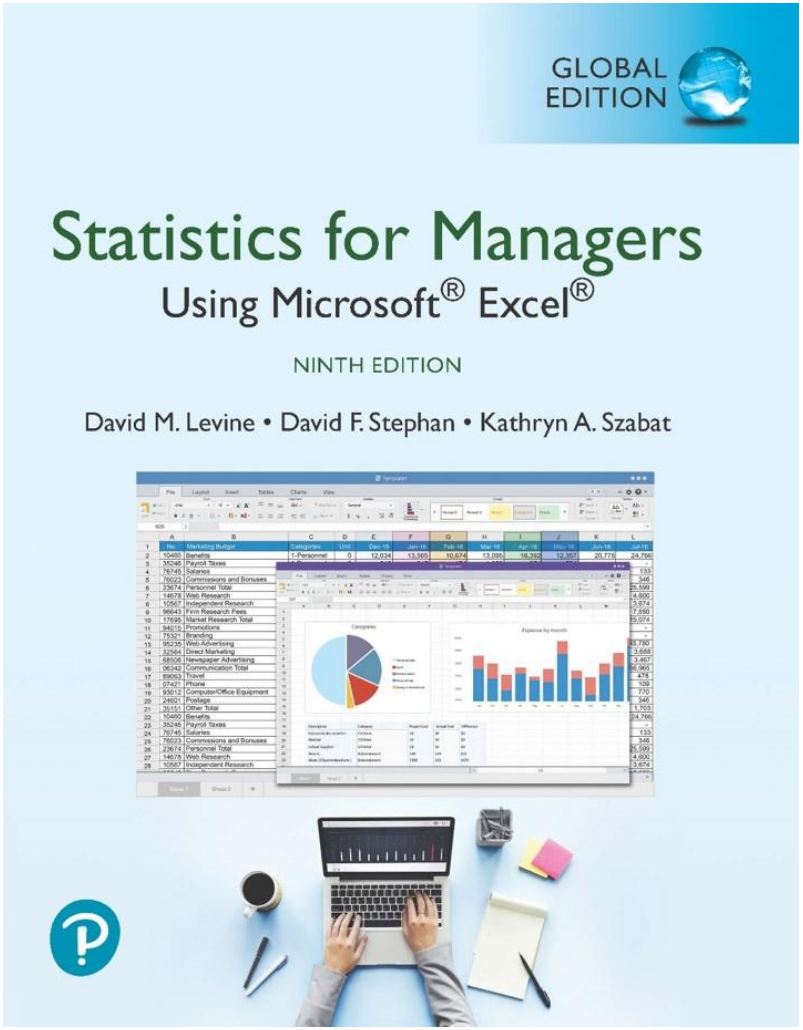


Chapter 10

Two-Sample Tests

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Objectives

In this chapter, you learn:

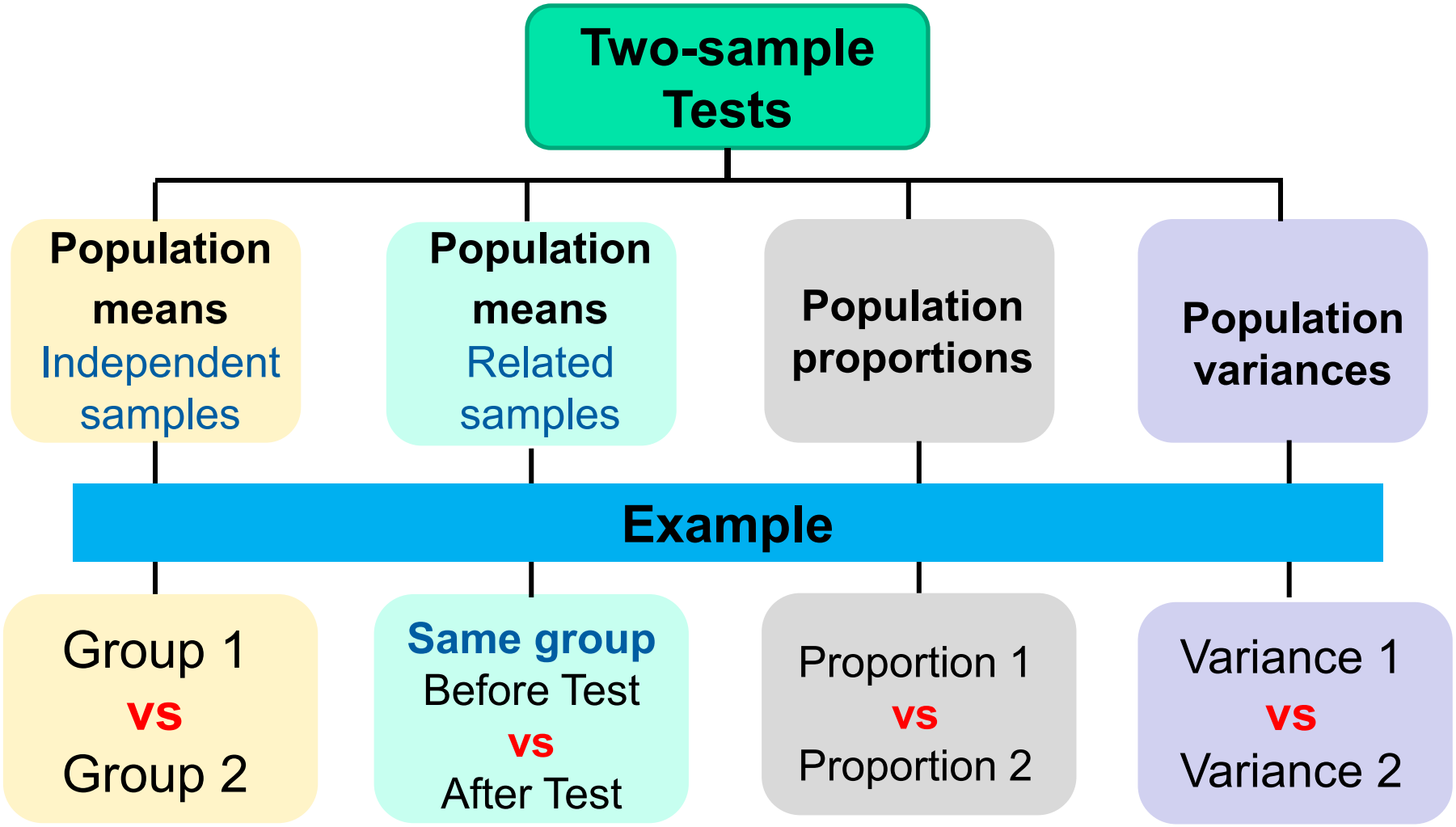
- How to compare the means or proportions of two independent populations.
- How to compare the means of two related populations.
- How to compare the variances of two independent populations.
- About the impact that sample size can have on statistical significance.

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Two-sample Tests



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Two Population Means

Independent Samples

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Two Population Means
Independent Samples

- **Goal:** Test the hypothesis or form a confidence interval for the difference between two population means, $\mu_1 - \mu_2$.
- **Assumption:**
  - Different data sources. That is, data sets are unrelated.
  - Data sets are randomly sampled and independent.
  - Sample selected from one population has no effect on the sample selected from the other population.
  - Populations are normally distributed or both sample sizes are at least 30.

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Two Population Means (Independent Samples)

Hypothesis Tests

Lower-tail test:

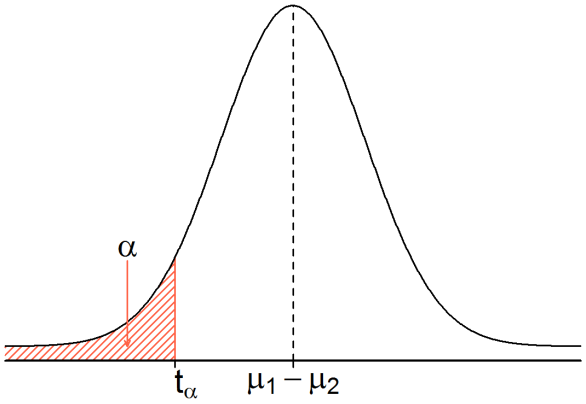
$\mathbf{H_0}: \mu_1 = \mu_2$

$\mathbf{H_1}: \mu_1 < \mu_2$

$\Downarrow$

$\mathbf{H_0}: \mu_1 - \mu_2 = 0$

$\mathbf{H_1}: \mu_1 - \mu_2 < 0$



Reject H_0 if $t_{stat} < t_\alpha$

Two-tail test:

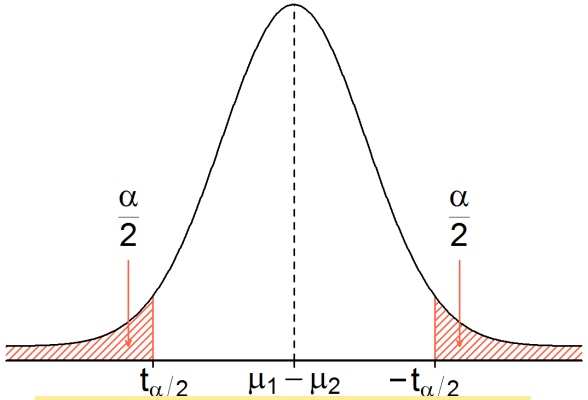
$\mathbf{H_0}: \mu_1 = \mu_2$

$\mathbf{H_1}: \mu_1 \neq \mu_2$

$\Downarrow$

$\mathbf{H_0}: \mu_1 - \mu_2 = 0$

$\mathbf{H_1}: \mu_1 - \mu_2 \neq 0$



Reject H_0 if $t_{stat} < t_{\alpha/2}$
or
Reject H_0 if $t_{stat} > -t_{\alpha/2}$

Upper-tail test:

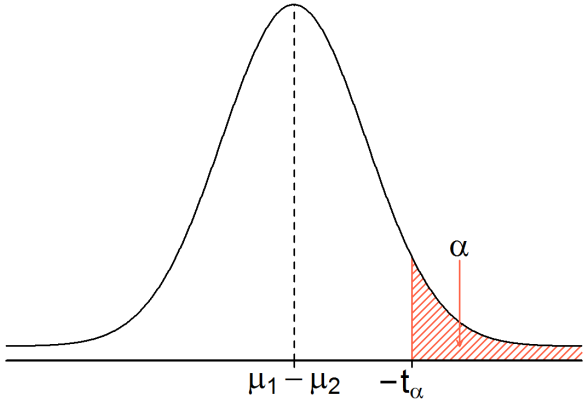
$\mathbf{H_0}: \mu_1 = \mu_2$

$\mathbf{H_1}: \mu_1 > \mu_2$

$\Downarrow$

$\mathbf{H_0}: \mu_1 - \mu_2 = 0$

$\mathbf{H_1}: \mu_1 - \mu_2 > 0$



Reject H_0 if $t_{stat} > -t_\alpha$

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Hypothesis Tests

- Pooled variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)}$$

- Test statistic:

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- Degrees of freedom (df) for t_{stat} is $(n_1 + n_2 - 2)$.

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Confidence Interval

- Confidence interval for $\mu_1 - \mu_2$ of $(1 - \alpha)$ level:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

- Degrees of freedom (df) for t_{stat} is $(n_1 + n_2 - 2)$.

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

You are comparing sales of a product from two different in-store locations: *Special-Up-Front* and *In-Aisle*. You have gathered the following monthly sales data.

PRODUCT LOCATION									
Special-Up-Front					**In-Aisle**				
224	189	248	285	273	192	236	164	154	189
190	243	215	280	317	220	261	186	219	202
	n_1		10			n_2		10	
	$\bar{x}_1$		246.4000			$\bar{x}_2$		202.3000	
	s_1^2		1809.8222			s_2^2		1058.0111	

Assuming both populations are approximately normally distributed with equal variances, is there a difference in mean sales ($\alpha = 0.05$)?

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

Calculate Test Statistic

$$\mathbf{H_0}: \ \mu_1 - \mu_2 = 0$$
$$\mathbf{H_1}: \ \mu_1 - \mu_2 \neq 0$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)} = \frac{9 \times 1809.8222 + 9 \times 1058.0111}{(10 - 1) + (10 - 1)} \approx 1433.9167$$

Therefore, test statistic is:

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(246.4 - 202.3) - 0}{\sqrt{1433.9167\left(\frac{1}{10} + \frac{1}{10}\right)}} \approx \mathbf{2.6041}$$

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

t test (critical value approach)

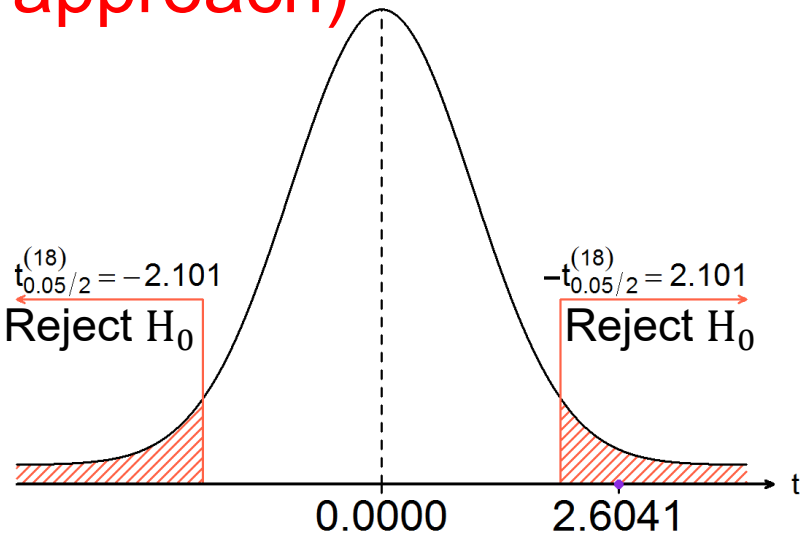
$$\mathbf{H_0}: \ \mu_1 - \mu_2 = 0$$
$$\mathbf{H_1}: \ \mu_1 - \mu_2 \neq 0$$

$\alpha = 0.05$

Degrees of freedom (df): $10 + 10 - 2 = 18$

Critical values: $\pm t_{\alpha/2}^{(\mathrm{df})} = \pm t_{0.025}^{(18)} = \pm 2.101$

Test statistic: $t_{stat} = 2.6041$



Decision: Reject H_0 at 5% significance level.

Conclusion: The mean sales differs significantly between the special-up-front and the in-aisle locations.

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

Confidence Interval for $(\mu_1 - \mu_2)$

$(1-\alpha) = (1-0.05) = 95\%$ confidence interval for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}^{(\mathrm{df})} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = (246.4 - 202.3) \pm t_{0.025}^{(18)} \sqrt{1433.9167 \left(\frac{1}{10} + \frac{1}{10}\right)}$$

$$= 44.1 \pm 2.101\sqrt{286.7833}$$

$$\approx \mathbf{(8.5202, 79.6798)}$$

Conclusion: The corresponding confidence interval does not contain zero. Also, both interval values are greater than zero. It is claimed that the mean sales at the special-up-front location is significantly greater than the mean sales at the in-aisle location.

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

t test (Excel): Critical value approach

	A	B	C	D	E	F
1	**Data**					
2	Special-Up-Front Sample (SUFS)			In-Aisle Sample (IAS)		
3	Sample Size		10	Sample Size		10
4	Sample Mean		246.4	Sample Mean		202.3
5	Sample Standard Deviation		42.5420	Sample Standard Deviation		32.5271
6				**Excel Commands**		
7	Hypothesis Test Details					
8	Hypothesised Difference			0		
9	Significance Level			0.05		
10						
11	Intermediate Calculations					
12	SUFS Degrees of Freedom			9	= C3 – 1	
13	IAS Degrees of Freedom			9	= F3 – 1	
14	Total Degrees of Freedom			18	= D12 + D13	
15	Pooled Variance			1433.9170	= ((D12 * C5^2) + (D13 * F5^2)) / D14	
16	Standard Error			16.9347	= SQRT(D15 * (1/C3 + 1/F3))	
17	Difference in Sample Means			44.1000	= C4 – F4	
18	t Test Statistic			2.6041	= (D17 – D8) / D16	
19						
20	Two-Tail Test					
21	Critical Value			2.1009	= T.INV(D9/2, D14)	
22						
23	Decision					
24	Reject the null hypothesis				= IF(ABS(D18)>D21, "Reject the null hypothesis", "Fail to reject the null hypothesis")	

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

t test (Excel): p-value approach

	A	B	C	D	E	F
1	**Data**					
2	Special-Up-Front Sample (SUFS)			In-Aisle Sample (IAS)		
3	Sample Size		10	Sample Size		10
4	Sample Mean		246.4	Sample Mean		202.3
5	Sample Standard Deviation		42.5420	Sample Standard Deviation		32.5271
6				**Excel Commands**		
7	Hypothesis Test Details					
8	Hypothesised Difference			0		
9	Significance Level			0.05		
10						
11	Intermediate Calculations					
12	SUFS Degrees of Freedom			9	= C3 – 1	
13	IAS Degrees of Freedom			9	= F3 – 1	
14	Total Degrees of Freedom			18	= D12 + D13	
15	Pooled Variance			1433.9170	= ((D12 * C5^2) + (D13 * F5^2)) / D14	
16	Standard Error			16.9347	= SQRT(D15 * (1/C3 + 1/F3))	
17	Difference in Sample Means			44.1000	= C4 – F4	
18	t Test Statistic			2.6041	= (D17 – D8) / D16	
19						
20	Two-Tail Test					
21	Critical Value			2.1009	= T.INV(D9/2, D14)	
22	p value			0.0179	= T.DIST.2T(ABS(D18), D14)	
23						
24	Decision					
25	Reject the null hypothesis				= IF(D22>D9, "Reject the null hypothesis", "Fail to reject the null hypothesis")	

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Two Population Means (Independent Samples)

$\sigma_1 = \sigma_2$ and are unknown ■ Example

Confidence Interval for $\mu_1 - \mu_2$ (Excel)

	A	B	C	D	E	F
1	**Data**					
2	Special-Up-Front Sample (SUFS)			In-Aisle Sample (IAS)		
3	Sample Size		10	Sample Size		10
4	Sample Mean		246.4	Sample Mean		202.3
5	Sample Standard Deviation		42.5420	Sample Standard Deviation		32.5271
6				**Excel Commands**		
7	Hypothesis Test Details					
8	Hypothesised Difference			0		
9	Significance Level			0.05		
10						
11	Intermediate Calculations					
12	SUFS Degrees of Freedom			9	= C3 – 1	
13	IAS Degrees of Freedom			9	= F3 – 1	
14	Total Degrees of Freedom			18	= D12 + D13	
15	Pooled Variance			1433.9170	= ((D12 * C5^2) + (D13 * F5^2)) / D14	
16	Standard Error			16.9347	= SQRT(D15 * (1/C3 + 1/F3))	
17	Difference in Sample Means			44.1000	= C4 – F4	
18	t Test Statistic			2.6041	= (D17 – D8) / D16	
19						
20	Two-Tail Test					
21	Critical Value			2.1009	=T.INV(D9/2, D14)	
22	p value			0.0179	= T.DIST.2T(ABS(D18), D14)	
23						
24	95% Confidence Interval					
25	Lower Limit			8.5216	= D17 - D21 * D16	
26	Upper Limit			79.6784	= D17 + D21 * D16	

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Two Population Means (Independent Samples)

$\sigma_1 \neq \sigma_2$ and are unknown

- Population variance are unknown and cannot be assumed to be equal.
- The corresponding test is known as **unequal variance t test**.
- The unequal variance t test is beyond the scope of this course. Therefore, it will **not** be discussed any further.

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Two Population Means

Related Samples

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Two Population Means (Related Samples)
The Paired Difference Test

- Tests means of two related populations.
  - Paired or matched samples.
  - Repeated measures (before/after treatment).
  - Use difference between paired values.
- Eliminates variation among experimental units.
- Assumptions:
  - Differences are normally distributed.
  - For non-normally distributed differences, sample size is at least 30 such that central limit theorem is applicable.

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Two Population Means (Related Samples)

The Paired Difference Test

- Consider two paired variables, $\mathbf{X_1}$ and $\mathbf{X_2}$, that each has sample size $n_1 = n_2 = \boldsymbol{n}$.
- The paired difference between observation i of both variables is:

$$\boldsymbol{d_i = x_{1i} - x_{2i}}$$

- The sample estimates of the mean $(\bar{x}_d)$ and the standard deviation (s_d) of the paired differences are:

$$\bar{\boldsymbol{d}} = \frac{\boldsymbol{1}}{\boldsymbol{n}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \boldsymbol{d_i}$$

$$\boldsymbol{s_d} = \sqrt{\frac{\boldsymbol{1}}{\boldsymbol{n-1}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \left(\boldsymbol{d_i - \bar{d}}\right)^{\boldsymbol{2}}}$$

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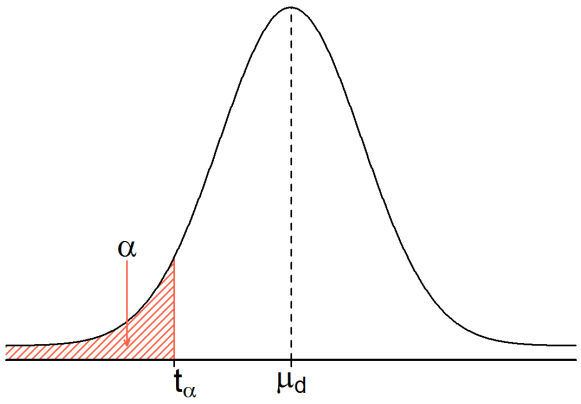
Two Population Means (Related Samples)

Hypothesis Tests

Lower-tail test:

$\mathbf{H_0}$: $\mu_d = 0$

$\mathbf{H_1}$: $\mu_d < 0$

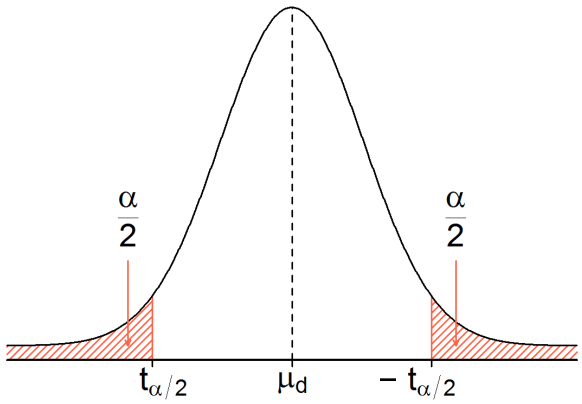


Reject H_0 if $t_{stat} < t_\alpha$

Two-tail test:

$\mathbf{H_0}$: $\mu_d = 0$

$\mathbf{H_1}$: $\mu_d \neq 0$

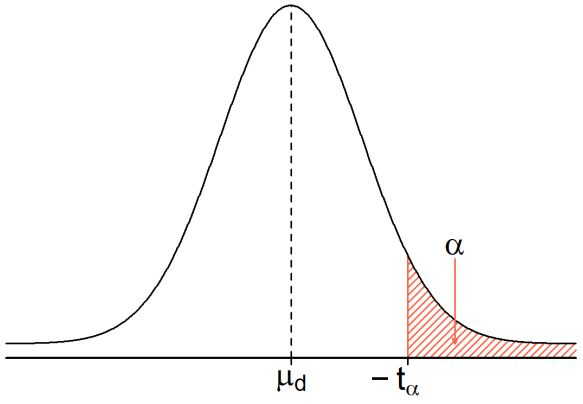


Reject H_0 if $t_{stat} < t_{\alpha/2}$
or
Reject H_0 if $t_{stat} > -t_{\alpha/2}$

Upper-tail test:

$\mathbf{H_0}$: $\mu_d = 0$

$\mathbf{H_1}$: $\mu_d > 0$



Reject H_0 if $t_{stat} > -t_\alpha$

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Two Population Means (Related Samples)
The Paired Difference Test

- Test statistic:

$$t_{stat} = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$$

- Confidence interval for μ_d at $(1 - \alpha) \times 100\%$ level:

$$\bar{d} \pm t_{\alpha/2} \times \frac{s_d}{\sqrt{n}}$$

- Degrees of freedom (df) for t_{stat} is $n - 1$.

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Two Population Means (Related Samples)

The Paired Difference Test: Example

A researcher wants to determine if there is any difference between the mean price of a market basket of $n = 7$ items at Checkers and Pick n Pay. The data is presented below.

Assume that prices of goods are normally distributed.

Item	Checkers	Pick n Pay	Difference
Chicken Broth	5.98	5.88	0.10
Ice Cream	8.59	7.19	1.40
Dishwasher Detergent	9.00	17.00	-8.00
Laundry Detergent	11.00	12.00	-1.00
Paper Towels	1.47	2.09	-0.62
Toilet Paper	12.00	27.00	-15.00
Facial Tissues	1.23	1.12	0.11

a. Is there difference in the mean of item prices at $\alpha = 0.05$?

b. Construct a 95% confidence interval for the population mean of item price differences.

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Two Population Means (Related Samples)

The Paired Difference Test: Example

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i = \frac{1}{7}\sum_{i=1}^{7}(0.10 + 1.40 + \cdots + 0.11) \approx \mathbf{-3.2871}$$

$$s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(d_i - \bar{d}\right)^2} = \sqrt{\frac{1}{7-1}\sum_{i=1}^{7}[(0.10 + 3.2781)^2 + \cdots]} \approx \mathbf{6.0101}$$

$$t_{stat} = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}} = \frac{-3.2781 - 0}{6.0101/\sqrt{7}} \approx \mathbf{-1.4471}$$

$\mathbf{H_0}$: $\mu_d = 0$

$\mathbf{H_1}$: $\mu_d \neq 0$

Significance level $(\alpha) = 0.05$

Degrees of freedom (df) $= n - 1 = 7 - 1 = 6$

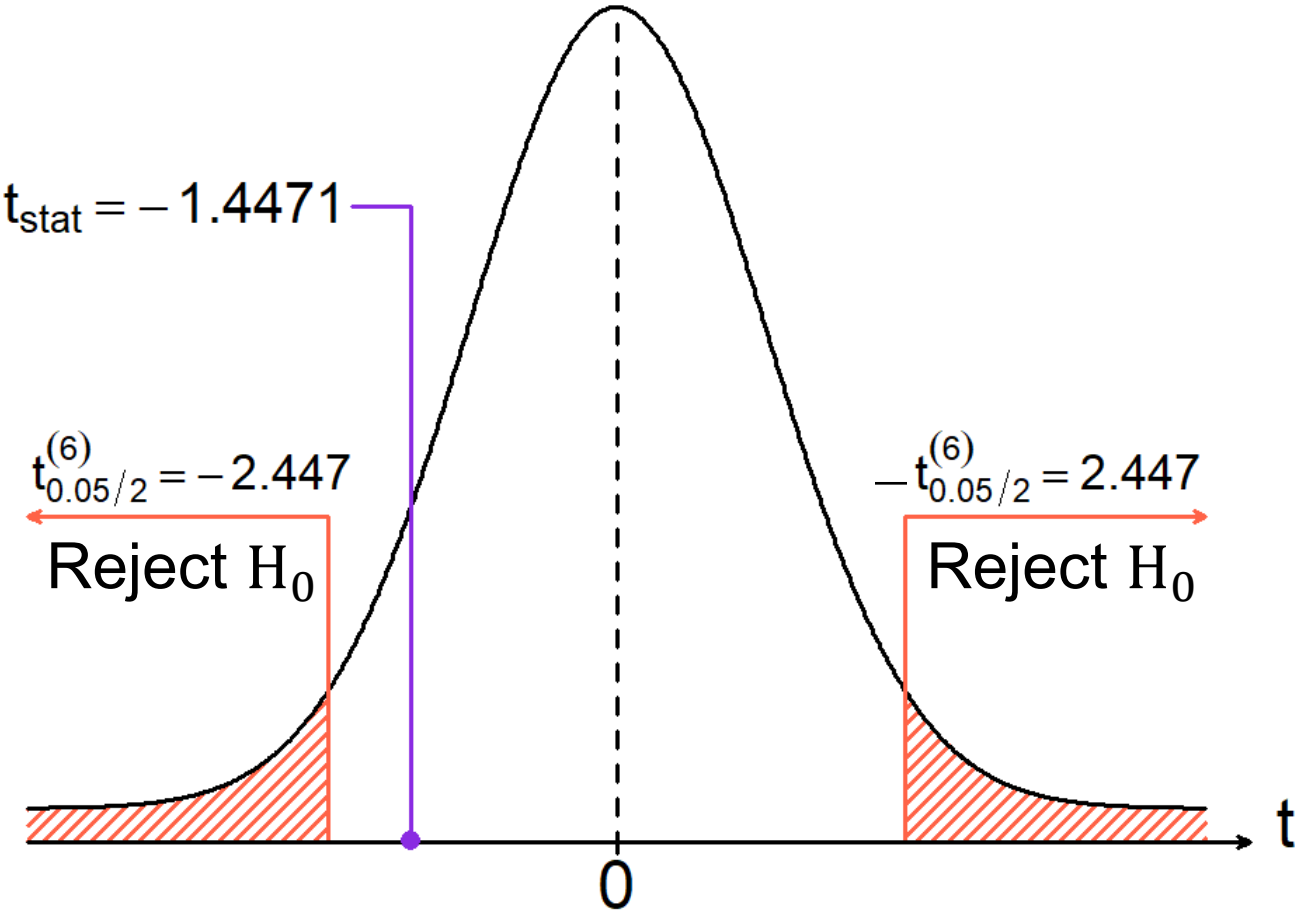
$$t_{\alpha/2}^{(\mathrm{df})} = t_{0.05/2}^{(6)} = \mathbf{-2.447}$$

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Two Population Means (Related Samples)

The Paired Difference Test: Example



Decision: $t^{(6)}_{0.05/2} = -2.447 < t_{stat} = -1.4471$. Therefore, fail to reject H_0.

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Two Population Means (Related Samples)
The Paired Difference Test: Example

95% confidence interval for μ_d:

$$\bar{d} \pm t_{\alpha/2}^{(\mathrm{df})} \times \frac{s_d}{\sqrt{n}} = -3.2871 \pm 2.4469 \times \frac{6.0101}{\sqrt{7}} = [\mathbf{-8.8456; 2.2713}]$$

Decision: The 95% confidence interval contains the null hypothesis value (that is, zero). Therefore, fail to reject H_0.

Conclusion: The sample data does not contain enough evidence for the null hypothesis to be rejected. It is therefore claimed that there is no statistically significant difference between the mean price of a basket of items from Checkers and Pick n Pay.

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Two Population Means (Related Samples)

The Paired Difference Test (Excel): Example

	A	B	C	D
1	**Data**			**Excel Commands**
2	Sample Size		**7**	
3	Average of the Differences		**-3.2871**	
4	Standard Deviation of the Differences		**6.0101**	
5				
6	**Hypothesis Test Details**			
7	Hypothesised Mean Difference		**0.00**	
8	Significance Level		**0.05**	
9				
10	**Intermediate Calculations**			
11	Degrees of Freedom		**6**	= C2 – 1
12	Standard Error		**2.2716**	= C4 / SQRT(C2)
13	t Test Statistic		**-1.4470**	= (C3 - C7) / C12
14				
15	**Two-Tail Test (Critical Value Approach)**			
16	Critical Value		**2.4469**	= T.INV.2T(C8, C11)
17	Decision:			
18		Fail to reject null hypothesis		= IF(ABS(C13)>C16, "Reject null hypothesis", "Fail to reject null hypothesis")
19				
20	**Two-Tail Test (p Value Approach)**			
21	p Value		**0.1980**	= T.DIST.2T(ABS(C13), C11)
22	Decision:			
23		Fail to reject null hypothesis		= IF(C21<C8, "Reject null hypothesis", "Fail to reject null hypothesis")

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Two Population Proportions

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Two Population Proportions

Notation

- n_1: sample size taken from population 1.
- n_2: sample size taken from population 2.
- x_1: count of items of interest from sample 1.
- x_2: count of items of interest from sample 2.
- π_1: population proportion value for population 1.
- π_2: population proportion value for population 2.
- $p_1 = x_1/n_1$: estimate of proportion obtained from sample 1.
- $p_2 = x_2/n_2$: estimate of proportion obtained from sample 2.
- $\bar{p} = \frac{x_1+x_2}{n_1+n_2}$: pooled estimate for the overall proportion.

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Two Population Proportions

- **Goal:**
  - Test a hypothesis or form a confidence interval for the difference between two population proportions.

$$\boldsymbol{\pi_1 - \pi_2}$$

- **Assumptions:**

$$n_1\pi_1 \geq 5 \quad \text{and} \quad n_1(1-\pi_1) \geq 5$$
$$n_2\pi_2 \geq 5 \quad \text{and} \quad n_2(1-\pi_2) \geq 5$$

- **Test statistic:**

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

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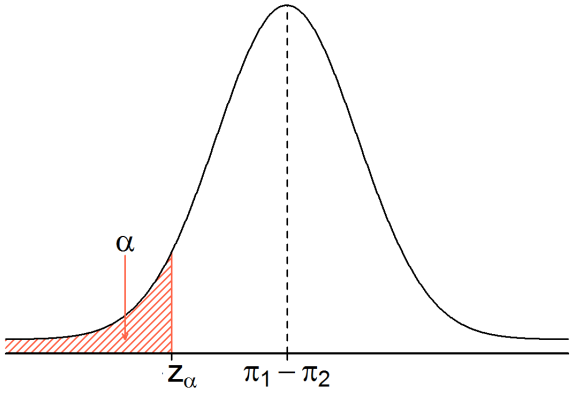
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Two Population Proportions

Lower-tail test:

$\mathbf{H_0}$: $\pi_1 - \pi_2 = 0$

$\mathbf{H_1}$: $\pi_1 - \pi_2 < 0$

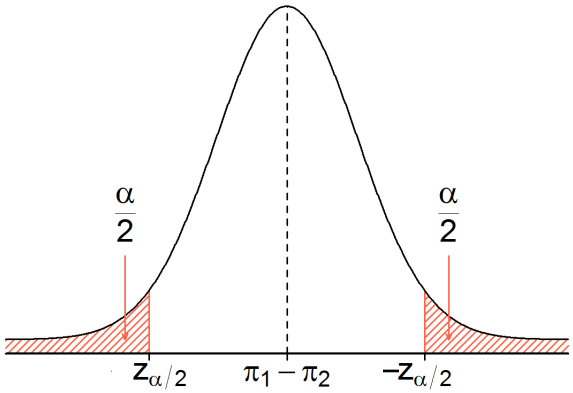


Reject H_0 if $z_{stat} < z_\alpha$

Two-tail test:

$\mathbf{H_0}$: $\pi_1 - \pi_2 = 0$

$\mathbf{H_1}$: $\pi_1 - \pi_2 \neq 0$

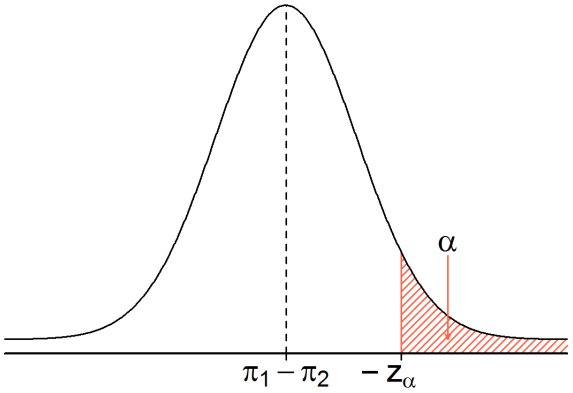


Reject H_0 if $z_{stat} < z_{\alpha/2}$
or
Reject H_0 if $z_{stat} > -z_{\alpha/2}$

Upper-tail test:

$\mathbf{H_0}$: $\pi_1 - \pi_2 = 0$

$\mathbf{H_1}$: $\pi_1 - \pi_2 > 0$



Reject H_0 if $z_{stat} > -z_\alpha$

Confidence Interval for $\pi_1 - \pi_2$

$$(p_1 - p_2) \pm z_{\alpha/2}\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

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Two Population Proportions

Example

In an experiment to compare two webpage designs, 387 of 4325 visitors signed up from Design 1 while 283 of 4639 visitors signed up from Design 2. At the significance level of $\alpha = 5\%$,

a. test if there is a difference between the proportion of sign-ups generated by the two different webpage designs?

b. construct a $(1 - \alpha) \times 100\%$ confidence interval for the difference between the two population proportions of sign-ups.

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Two Population Proportions

Example

- **Sample proportions:**

$$\text{Design 1:} \quad p_1 \approx \frac{387}{4325} \approx \mathbf{0.0895}$$

$$\text{Design 2:} \quad p_2 \approx \frac{283}{4639} \approx \mathbf{0.0610}$$

- **Pooled estimate for overall proportion:**

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{387 + 283}{4325 + 4639} \approx \mathbf{0.0747}$$

$$\mathbf{H_0}: \ \pi_1 - \pi_2 = 0$$
$$\mathbf{H_1}: \ \pi_1 - \pi_2 \neq 0$$

- **Test statistic for $\boldsymbol{\pi_1 - \pi_2}$:**

$$z_{stat} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.0895 - 0.0610) - 0}{\sqrt{0.0747(1-0.0747)\left(\frac{1}{4325} + \frac{1}{4639}\right)}} \approx \mathbf{5.1228}$$

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Two Population Proportions

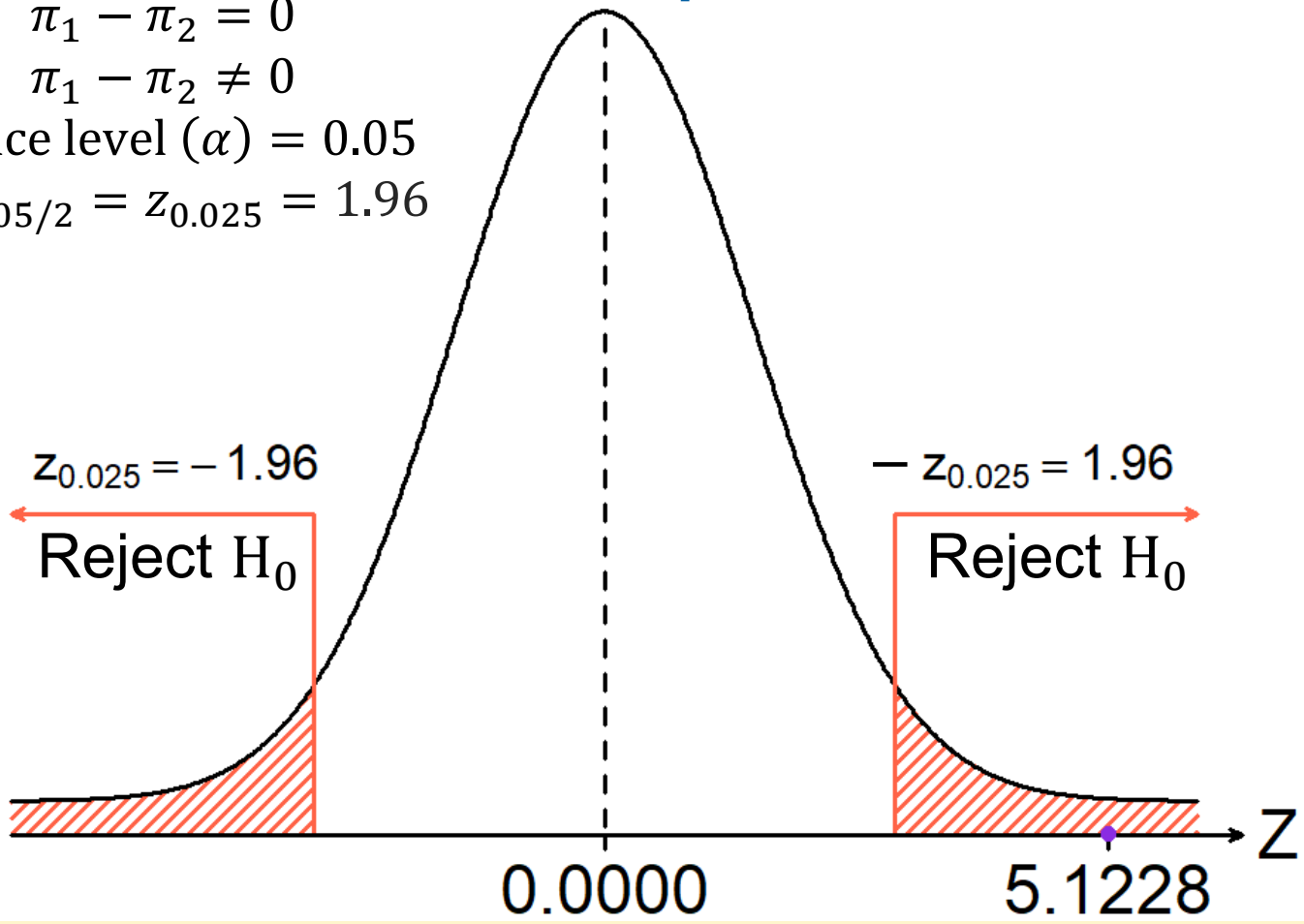
Example

$\mathbf{H_0}$: $\pi_1 - \pi_2 = 0$

$\mathbf{H_1}$: $\pi_1 - \pi_2 \neq 0$

Significance level $(\alpha) = 0.05$

$z_{\alpha/2} = z_{0.05/2} = z_{0.025} = 1.96$



Decision: $z_{stat} = 5.1228 > -z_{0.025} = 1.96$. Therefore, reject H_0.

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Two Population Proportions

Example

- **95% Confidence interval for $\pi_1 - \pi_2$:**

$$(p_1 - p_2) \pm z_{0.025}\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} =$$

$$(0.0895 - 0.0610) \pm 1.96\sqrt{\frac{0.0895(1-0.0895)}{4325} + \frac{0.0610(1-0.0610)}{4639}} = \mathbf{(0.0175; 0.0394)}$$

Decision: The 95% confidence interval does **not** contain the null hypothesis value (that is, zero). Thus, reject H_0.

Conclusion: There is evidence of a statistically significant difference between the proportions of sign-ups from the two webpage designs. Design 1 tend to attract more sign-ups.

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Two Population Proportions

Example (Excel)

	A	B	C
1	**Data**		**Excel Commands**
2	Design 1: Sign-ups	387	
3	Design 1: Visits	4325	
4	Design 2: Sign-ups	283	
5	Design 2: Visits	4639	
6			
7	Hypothesis Test Details		
8	Hypothesised Difference	0.00	
9	Significance Level	0.05	
10			
11	Intermediate Calculations		
12	Design 1: Sample Proportion	0.0895	= B2 / B3
13	Design 2: Sample Proportion	0.0610	= B4 / B5
14	Difference Between Sample Proportions	0.0285	= B12 - B13
15	Average Proportion	0.0747	= (B2 + B4) / (B3 + B5)
16	Test Statistic Denominator	0.0056	= SQRT(B15 * (1 - B15) * (1/B3 + 1/B5))
17	z Test Statistic	5.1228	= (B14 - B8) / B16
18			
19	Two-Tail Test (Critical Value Approach)		
20	Critical Value	1.9600	= ABS(NORM.S.INV(B9/2))
21	Decision:	Reject H0	= IF(ABS(B17)>B20, "Reject H0", "Fail to reject H0")
22			
23	Two-Tail Test (p Value Approach)		
24	p Value	0.0000	= 2 * (1 - NORM.S.DIST(ABS(B17), TRUE))
25	Decision:	Reject H0	= IF(B24<B9, "Reject H0", "Fail to reject H0")

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Ratio of Two Variances

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Testing for the Ratio of Two Population Variances

- **Goal:**
  - Test a hypothesis about the ratio of two population variances.

$$\boldsymbol{\sigma_1^2 \div \sigma_2^2}$$

- **Test statistic:**

$$\mathrm{F}_{stat} = \frac{s_1^2}{s_2^2}$$

  - $\boldsymbol{s_1^2}$: larger variance estimate (always at the numerator).
  - $\boldsymbol{s_2^2}$: smaller variance estimate (always at the denominator).
  - This order of the sample variances will ensure $\mathrm{F}_{stat} \geq 1$ always.
- **Degrees of freedom $(\mathbf{df})$:**
  - $\mathbf{df_1 = n_1 - 1}$ where n_1 is the size of the sample that generated s_1^2.
  - $\mathbf{df_2 = n_2 - 1}$ where n_2 is the size of the sample that generated s_2^2

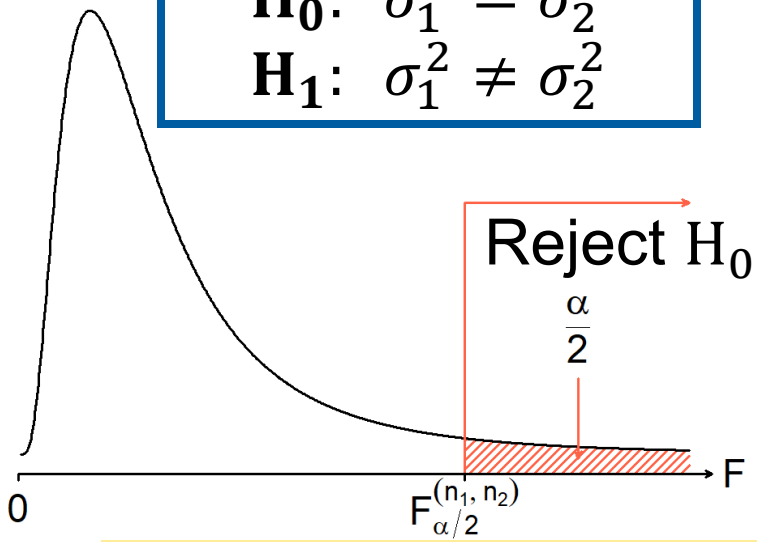
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Testing for the Ratio of Two Population Variances

Two-tail test:

$\mathbf{H_0}$: $\sigma_1^2 = \sigma_2^2$

$\mathbf{H_1}$: $\sigma_1^2 \neq \sigma_2^2$

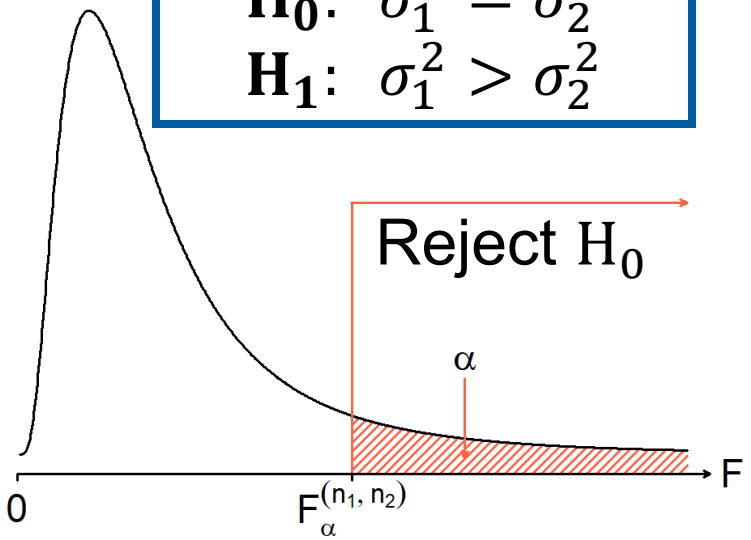


Reject H_0 if $F_{stat} > F_{\alpha/2}^{(df_1, df_2)}$

Upper-tail test:

$\mathbf{H_0}$: $\sigma_1^2 = \sigma_2^2$

$\mathbf{H_1}$: $\sigma_1^2 > \sigma_2^2$



Reject H_0 if $F_{stat} > F_{\alpha}^{(df_1, df_2)}$

The $F_{\alpha}^{(df_1, df_2)}$ Critical Value from the F table

- $\mathbf{df_1}$: numerator degree of freedom.
- $\mathbf{df_2}$: denominator degree of freedom.

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Testing for the Ratio of Two Population Variances

Example

Test the hypothesis (at a 5% significance level) that the variability in number of sales at two different stores is similar, given the following sample information.

Item	Store A	Store B
Sample size	10	10
Sample standard deviation	32.5271	42.5420

Since $s_B > s_A$ therefore,

$$s_1 = s_B \quad \text{and} \quad s_2 = s_A$$

$$df_1 = n_B - 1 = 10 - 1 = \mathbf{9} \quad \text{and} \quad df_2 = n_A - 1 = 10 - 1 = \mathbf{9}$$

$$F_{\alpha/2}^{(df_1, df_2)} = F_{0.025}^{(9,9)} = \mathbf{4.03}$$

$$F_{stat} = \frac{s_1^2}{s_2^2} = \frac{42.5420^2}{32.5271^2} \approx \mathbf{1.7106}$$

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Testing for the Ratio of Two Population Variances

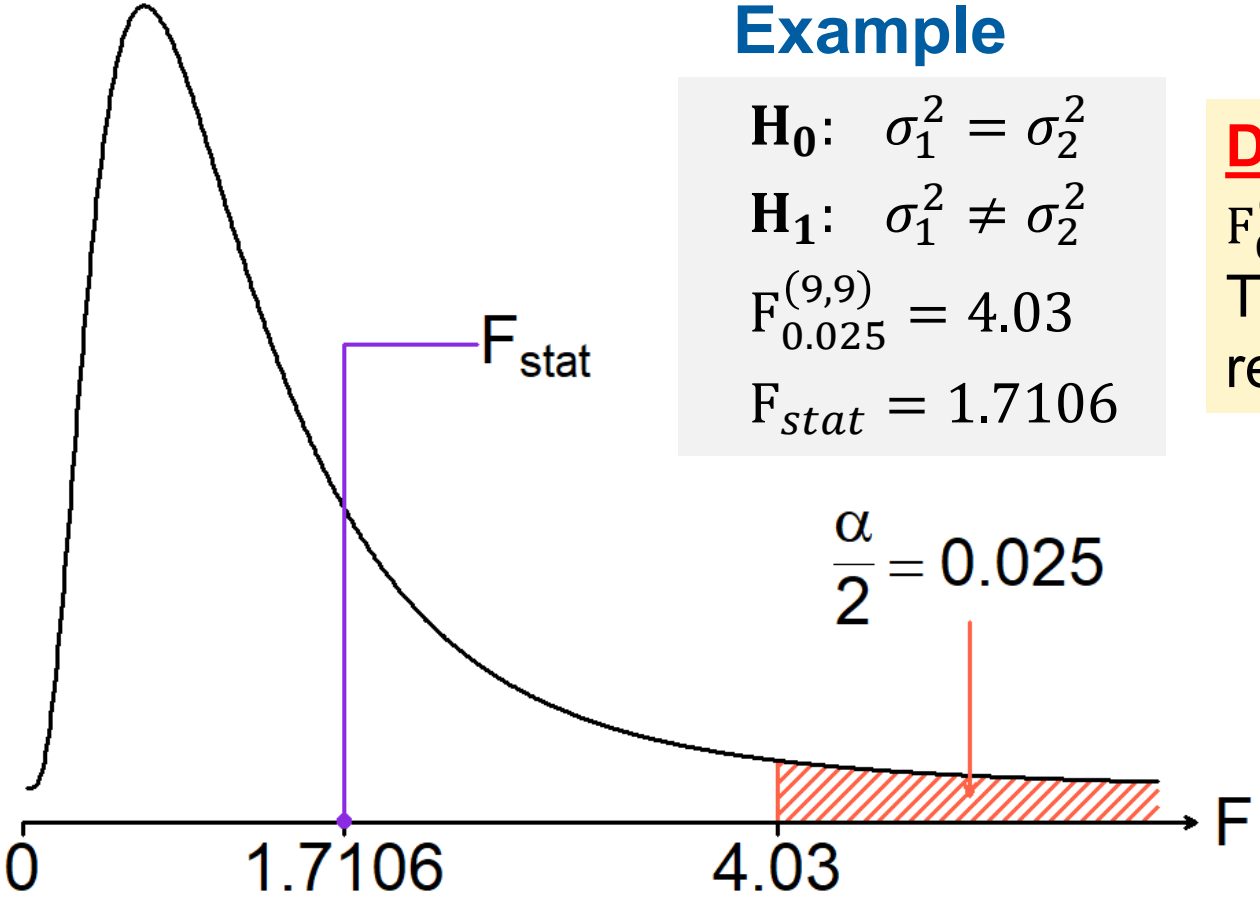
Example

$\mathbf{H_0}$: $\sigma_1^2 = \sigma_2^2$

$\mathbf{H_1}$: $\sigma_1^2 \neq \sigma_2^2$

$F_{0.025}^{(9,9)} = 4.03$

$F_{stat} = 1.7106$



Decision:

$F_{0.025}^{(9,9)} > F_{stat}$.

Therefore, fail to reject H_0.

Conclusion: There is not evidence of a statistically significant difference between the variation of sales in the two stores.

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Testing for the Ratio of Two Population Variances

Example (Excel)

	A	B	C
1	**Data**		**Excel Commands**
2	Store A: Sample Size	10	
3	Store B: Sample Size	10	
4	Store A: Standard Deviation (Smaller)	32.5271	
5	Store B: Standard Deviation (Larger)	42.5420	
6			
7	**Hypothesis Test Details**		
8	Significance Level	0.05	
9			
10	**Intermediate Calculations**		
11	Store A: Degrees of Freedom	9	= B2 – 1
12	Store B: Degrees of Freedom	9	= B3 – 1
13	F Test Statistic	1.7106	= (B5^2) / (B4^2)
14			
15	**Two-Tail F Test (Critical Value Approach)**		
16	Critical Value	4.0260	= F.INV.RT(B8/2, B12, B11)
17	Decision:	Fail to reject H0	= IF(ABS(B13)>B16, "Reject H0", "Fail to reject H0")
18			
19	**Two-Tail F Test (p Value Approach)**		
20	p Value	0.4361	= 2 * F.DIST.RT(B13, B12, B11)
21	Decision:	Fail to reject H0	= IF(B20<B8, "Reject H0", "Fail to reject H0")

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Effect Size

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Effect Size

Motivational Example

In an investigation about wages, the hourly earnings of workers at two factories were collected. See below for data summary. Assuming both population wages are approximately normally distributed with equal variances, is there a difference in mean sales yield $(\alpha = 0.05)$?

a) If the provided estimates were obtained from sample sizes of ten employee wages from each factory.

b) If sample sizes of twenty employee wages at each of the factories generated the provided estimates.

	Factory A	Factory B
Sample mean $(\bar{x})$	R29.48	R25.95
Sample standard deviation (s)	R3.50	R5.16

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Effect Size

Motivational Example

$\mathbf{H_0}$: $\mu_1 - \mu_2 = 0$ $\mathbf{H_1}$: $\mu_1 - \mu_2 \neq 0$

Case A:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)} = \frac{9 \times 3.50^2 + 9 \times 5.16^2}{(10 - 1) + (10 - 1)} \approx 19.4378$$

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(29.48 - 25.95) - 0}{\sqrt{19.4378\left(\frac{1}{10} + \frac{1}{10}\right)}} \approx \mathbf{1.7903}$$

Case B:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 - 1) + (n_2 - 1)} = \frac{19 \times 3.50^2 + 19 \times 5.16^2}{(20 - 1) + (20 - 1)} \approx 19.4378$$

$$t_{stat} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(29.48 - 25.95) - 0}{\sqrt{19.4378\left(\frac{1}{20} + \frac{1}{20}\right)}} \approx \mathbf{2.5319}$$

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Effect Size

Motivational Example

$\mathbf{H_0}$: $\mu_1 - \mu_2 = 0$ $\quad\quad$ $\mathbf{H_1}$: $\mu_1 - \mu_2 \neq 0$

$$\alpha = 0.05$$

Case A:

$\mathrm{df} = 10 + 10 - 2 = 18$

$\pm t_{\alpha/2}^{(\mathrm{df})} = \pm t_{0.025}^{(18)} = \pm 2.101$

$t_{stat} = 1.7903$

Decision: Fail to reject H_0 since $t_{stat} < t_{0.025}^{(18)}$.

Case B:

$\mathrm{df} = 20 + 20 - 2 = 38$

$\pm t_{\alpha/2}^{(\mathrm{df})} = \pm t_{0.025}^{(38)} \approx \pm 2.021$

$t_{stat} = 2.5319$

Decision: Reject H_0 since $t_{stat} > t_{0.025}^{(38)}$.

- **NOTE**: the difference in sample sizes produced different decisions!. Thus, there is need to account for the effect of sample size on hypothesis test results.

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Effect Size (not covered in detail)

- Availability of big data increases the likelihood of finding statistically significant but practically unimportant results.
- Need to find a way to measure the importance of a result and not just its statistical significance.
- One method to accomplish this is to measure the **effect size**.
- Can use confidence interval to describe the uncertainty in the size of the result.
- **Standardised effect size:** a standard way to classify effect size
  - Standardised effect size $\approx 0.20 \Rightarrow$ **small**.
  - Standardised effect size $\approx 0.50 \Rightarrow$ **medium**.
  - Standardised effect size $\geq 0.80 \Rightarrow$ **large**.

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Chapter Summary

In this chapter, we discussed:

- Comparing the means or proportions of two independent populations.
- Comparing the means of two related populations.
- Comparing the variances of two independent populations.
- How to interpret standardised effect size of the test.

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